

# Recitation 2

Causal Inference, Confounding, and Simpson's Paradox  
Solutions and Teaching Notes

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## Pacing guide (75 minutes)

| Time      | Activity             | Suggested use                                      |
|-----------|----------------------|----------------------------------------------------|
| 0–25 min  | Review               | Potential outcomes, confounding, Curry–Lee example |
| 25–38 min | Problem 1            | 7 min individual work, then discussion             |
| 38–53 min | Problem 2            | 8 min individual work, then discussion             |
| 53–68 min | Problem 3            | 8 min individual work, then discussion             |
| 68–75 min | Wrap-up or Problem 4 | Use Problem 4 only if the class is ahead           |

**Teaching emphasis.** Keep three quantities distinct: an observed conditional probability, a potential-outcome probability, and a standardized probability. In the basketball example, emphasize that standardization is a better descriptive comparison but is causal only under the stated conditional-independence assumption.

## 1 Potential outcomes and causal effects

Let the binary treatment  $\tilde{t}$  equal 1 for treatment and 0 for control. Every unit has two **potential outcomes**:

$$\widetilde{po}_0 = \text{outcome under control}, \quad \widetilde{po}_1 = \text{outcome under treatment}.$$

Both are defined whether or not the unit actually receives treatment. The observed outcome is

$$\tilde{y} = \begin{cases} \widetilde{po}_0, & \tilde{t} = 0, \\ \widetilde{po}_1, & \tilde{t} = 1, \end{cases} \quad \text{or equivalently} \quad \tilde{y} = (1 - \tilde{t})\widetilde{po}_0 + \tilde{t}\widetilde{po}_1.$$

This relationship is called **consistency**. For one unit we observe only one potential outcome; the other is a **counterfactual**. This is the fundamental problem of causal inference.

For a binary outcome, the population causal risk difference is

$$\Delta_{\text{causal}} := \mathbb{P}(\widetilde{po}_1 = 1) - \mathbb{P}(\widetilde{po}_0 = 1) = \mathbb{E}[\widetilde{po}_1] - \mathbb{E}[\widetilde{po}_0].$$

The difference visible in observational data is instead

$$\Delta_{\text{obs}} := \mathbb{P}(\tilde{y} = 1 \mid \tilde{t} = 1) - \mathbb{P}(\tilde{y} = 1 \mid \tilde{t} = 0).$$

By consistency, the first term describes  $\widetilde{po}_1$  among treated units and the second describes  $\widetilde{po}_0$  among control units. These are generally different subpopulations, so association need not equal causation.

If treatment is independent of both potential outcomes,

$$\tilde{t} \perp\!\!\!\perp \{\widetilde{p}o_0, \widetilde{p}o_1\},$$

then  $\Delta_{\text{obs}} = \Delta_{\text{causal}}$  at the population level. Random assignment creates this independence by design. In a finite randomized sample, the two observed rates are still estimates and need not equal the population probabilities exactly.

## 2 Confounding and adjustment

A pre-treatment variable  $\tilde{a}$  is a **confounder** when it helps create dependence between treatment and the potential outcomes. For example, age may affect which patients take a drug and may also predict recovery under either treatment. Comparing treated and control patients without accounting for age then mixes together treatment differences and age-composition differences.

Suppose the following assumptions are plausible:

1. **Consistency:** the observed outcome equals the potential outcome under the treatment received.
2. **Conditional independence:**  $\{\widetilde{p}o_0, \widetilde{p}o_1\} \perp\!\!\!\perp \tilde{t} \mid \tilde{a}$ .
3. **Positivity:** within every relevant value of  $a$ , both treatment groups can occur.

Then, for  $r \in \{0, 1\}$ ,

$$\mathbb{P}(\widetilde{p}o_r = 1) = \sum_a \mathbb{P}(\tilde{a} = a) \mathbb{P}(\tilde{y} = 1 \mid \tilde{a} = a, \tilde{t} = r).$$

The same population weights  $\mathbb{P}(\tilde{a} = a)$  are used for both treatments. By contrast, the unadjusted observed rate is

$$\mathbb{P}(\tilde{y} = 1 \mid \tilde{t} = r) = \sum_a \mathbb{P}(\tilde{a} = a \mid \tilde{t} = r) \mathbb{P}(\tilde{y} = 1 \mid \tilde{a} = a, \tilde{t} = r),$$

which may use very different weights in the two treatment groups. This is the law of total probability, and the changing weights are the mechanism behind Simpson's paradox.

## 3 Simpson's paradox: Curry versus Lee

During the 2014–2015 NBA season, Courtney Lee had the higher overall three-point percentage, but Stephen Curry had the higher percentage within both distance categories:

| Shot type                      | Stephen Curry   | Courtney Lee   |
|--------------------------------|-----------------|----------------|
| Short threes ( $\leq 24$ feet) | 45/90 = 50.0%   | 56/116 = 48.3% |
| Long threes ( $> 24$ feet)     | 145/366 = 39.6% | 19/55 = 34.5%  |
| All threes                     | 190/456 = 41.7% | 75/171 = 43.9% |

This reversal is **Simpson's paradox**. Shot distance, denoted  $\tilde{d}$ , explains it. Lee took 116/171 = 67.8% of his attempts from the easier short range, while Curry took only 90/456 = 19.7% from that range. Each overall percentage is a weighted average with a different set of weights. For example,

$$\mathbb{P}(\tilde{y} = 1 \mid \tilde{t} = \text{Lee}) = 0.678(0.483) + 0.322(0.345) = 0.439.$$

For a common comparison, weight both players by the shot mix in the combined data:  $206/627 = 0.329$  short and  $421/627 = 0.671$  long. The standardized rates are

$$\text{Curry: } 0.329(0.500) + 0.671(0.396) = 0.430,$$

$$\text{Lee: } 0.329(0.483) + 0.671(0.345) = 0.390.$$

Standardization therefore favors Curry. It is a useful comparison of shooting under a common distance distribution. Interpreting it as a causal comparison requires the strong assumption that, conditional on distance, player identity is independent of the relevant potential outcomes; other shot characteristics may violate that assumption.

**Read the comparison before interpreting it.** Marginal rates answer “what happened under each group’s actual mix?” Stratum-specific rates answer “what happened within a fixed category?” Standardized rates answer “what would the rates be under a common mix?”

## Practice problems

### Problem 1: What is observed?

[7 minutes]

Four patients have binary treatment and recovery outcomes:

| Patient | Treatment $t$ | Observed $y$ | $po_0$ | $po_1$ |
|---------|---------------|--------------|--------|--------|
| A       | 0             | 0            |        |        |
| B       | 0             | 1            |        |        |
| C       | 1             | 1            |        |        |
| D       | 1             | 0            |        |        |

- (a) Fill in every potential outcome determined by the observed data. Mark each unobserved potential outcome with “?”.
- (b) Compute the observed difference in recovery rates, treated minus control.
- (c) Assuming only that potential outcomes are binary, find the smallest and largest possible finite-sample average causal effect  $\frac{1}{4} \sum_{i=1}^4 (po_{1,i} - po_{0,i})$ .
- (d) If treatment was randomly assigned, does that reveal each missing counterfactual? What does randomization provide instead?

**Solution.** Consistency determines the entries under the treatment actually received:

| Patient | $t$ | $y$ | $po_0$ | $po_1$ |
|---------|-----|-----|--------|--------|
| A       | 0   | 0   | 0      | ?      |
| B       | 0   | 1   | 1      | ?      |
| C       | 1   | 1   | ?      | 1      |
| D       | 1   | 0   | ?      | 0      |

Both groups recover at rate  $1/2$ , so the observed difference is 0.

The known  $po_1$  values sum to 1, and the two missing values can add anything from 0 to 2. Thus  $\sum_i po_{1,i} \in [1, 3]$ . The same argument gives  $\sum_i po_{0,i} \in [1, 3]$ . The difference of the sums can range from  $-2$  to  $2$ , so the finite-sample average causal effect can range from  $-1/2$  to  $1/2$ . The observed difference 0 does not determine it.

Randomization does not reveal any patient’s missing counterfactual. It makes treatment independent of potential outcomes, so group averages are unbiased estimators of the corresponding population potential-outcome means (and become accurate with sufficiently large samples).

**Problem 2: Diagnose the Curry–Lee reversal****[8 minutes]**

Use the basketball table in the review.

- (a) Which player has the higher marginal three-point percentage? Which player has the higher percentage within each distance category?
- (b) Write Curry’s overall percentage as a weighted average of his short- and long-shot percentages. Compare his weights with Lee’s.
- (c) In one or two sentences, explain why the direction reverses after the data are stratified by distance.
- (d) Using a target shot mix of 40% short and 60% long, compute a standardized percentage for each player. Which is larger?
- (e) Does part (d) by itself prove that Curry would outperform Lee on the same shots? State the additional assumption needed for that causal claim.

**Solution.** Lee has the higher marginal percentage: 43.9% versus 41.7%. Curry has the higher rate among both short attempts (50.0% versus 48.3%) and long attempts (39.6% versus 34.5%). Curry’s overall rate is

$$\frac{90}{456} \frac{45}{90} + \frac{366}{456} \frac{145}{366} = 0.197(0.500) + 0.803(0.396) = 0.417.$$

Lee’s weights are  $116/171 = 0.678$  short and  $55/171 = 0.322$  long. Lee’s overall rate receives much more weight from easier short attempts, which raises his marginal percentage even though his rate is lower within each category.

With the common 40/60 mix,

$$\text{Curry: } 0.4(0.500) + 0.6(0.396) = 0.4376,$$

$$\text{Lee: } 0.4(0.483) + 0.6(0.345) = 0.4002.$$

Curry’s standardized percentage is higher. A causal interpretation additionally requires the relevant potential outcomes to be independent of player identity conditional on distance. The table cannot establish this; other shot features may still confound the comparison.

**Problem 3: A drug and patient risk****[8 minutes]**

An observational study records the following recoveries:

| Patient group | Drug ( $t = 1$ ) | No drug ( $t = 0$ ) |
|---------------|------------------|---------------------|
| Lower risk    | 45/50 = 90%      | 80/100 = 80%        |
| Higher risk   | 30/100 = 30%     | 10/50 = 20%         |
| All patients  | 75/150 = 50%     | 90/150 = 60%        |

- Compute the unadjusted observed risk difference. What conclusion would it suggest?
- Compute the risk difference within each risk group. Describe the reversal.
- In the full sample, half the patients are lower risk and half are higher risk. Standardize both treatment groups to this common 50/50 mix. Compute the adjusted risk difference.
- Explain how risk group creates confounding here. State the assumptions needed to interpret the adjusted difference causally.

**Solution.** The unadjusted difference is

$$0.50 - 0.60 = -0.10,$$

which suggests that the drug lowers recovery by 10 percentage points. Within the lower-risk group the difference is  $0.90 - 0.80 = 0.10$ , and within the higher-risk group it is  $0.30 - 0.20 = 0.10$ . Thus the drug has the higher observed recovery rate in each stratum even though it has the lower marginal rate.

Using the same 50/50 weights for both treatment levels gives

$$\begin{aligned}\mathbb{P}(\widetilde{po}_1 = 1) &\approx 0.5(0.90) + 0.5(0.30) = 0.60, \\ \mathbb{P}(\widetilde{po}_0 = 1) &\approx 0.5(0.80) + 0.5(0.20) = 0.50.\end{aligned}$$

The adjusted risk difference is therefore 0.10.

Only one third of the treated group is lower risk (50/150), compared with two thirds of the untreated group (100/150). Risk group predicts recovery and is distributed differently across treatment groups, so the marginal comparison is confounded. The causal interpretation requires consistency, positivity, and conditional independence of treatment and both potential outcomes given risk group. It also assumes that the risk categories are measured well enough and that no important confounder remains unmeasured.

**Problem 4 (backup): True or false?**

*Optional if time allows. Suggested working time: 5–7 minutes.*

- Random assignment reveals both potential outcomes for every unit.
- If treatment has a positive risk difference in every stratum, then any standardized risk difference using the same nonnegative stratum weights for both treatments is positive.
- To adjust for a confounder, use the treated group's stratum weights for the treated rate and the control group's weights for the control rate.
- Seeing a Simpson reversal proves that one of the two marginal groups has been measured incorrectly.

**Solution.** The answers are false, true, false, false.

- (a) Randomization balances potential outcomes across treatment groups in distribution; it does not reveal a unit's counterfactual.
- (b) A common-weight standardized difference is a weighted average of the positive stratum-specific differences, so it is positive.
- (c) Those are exactly the treatment-specific weights used by the unadjusted marginal rates. Adjustment uses one common target distribution.
- (d) The reversal can arise from different stratum compositions, even when every count is correct.

## Takeaways

- Potential outcomes define causal questions, but only one is observed per unit.
- Randomization breaks dependence between treatment and potential outcomes.
- Confounding makes observed treatment groups incomparable; conditional comparisons and common-weight standardization can repair this only under substantive assumptions.
- Simpson's paradox is a weighted-average reversal, not an arithmetic error.

## Sources

The review follows the course handouts *Causal Inference* and *Modeling Multiple Discrete Variables*, together with [Was Courtney Lee Better Than Stephen Curry? Simpson's Paradox](#) from *Probability and Statistics for Data Science*.